10.1.2 . Reverse a String

A] Algorithm

Step 1: Start
Step 2: Read input string from the user and store it in user_input
Step 3: Define a function reverse_string(input_string)
Step 4: Inside the function, initialize an empty string re = ""
Step 5: Find the length l of input_string
Step 6: Loop from index l-1 down to 0 (reverse order):
    Step 6a: Append the character at current index to re
Step 7: After the loop, return re (the reversed string)
Step 8: Call reverse_string(user_input) and store the result in result
Step 9: Print the original string user_input
Step 10: Print the reversed string result
Step 11: End

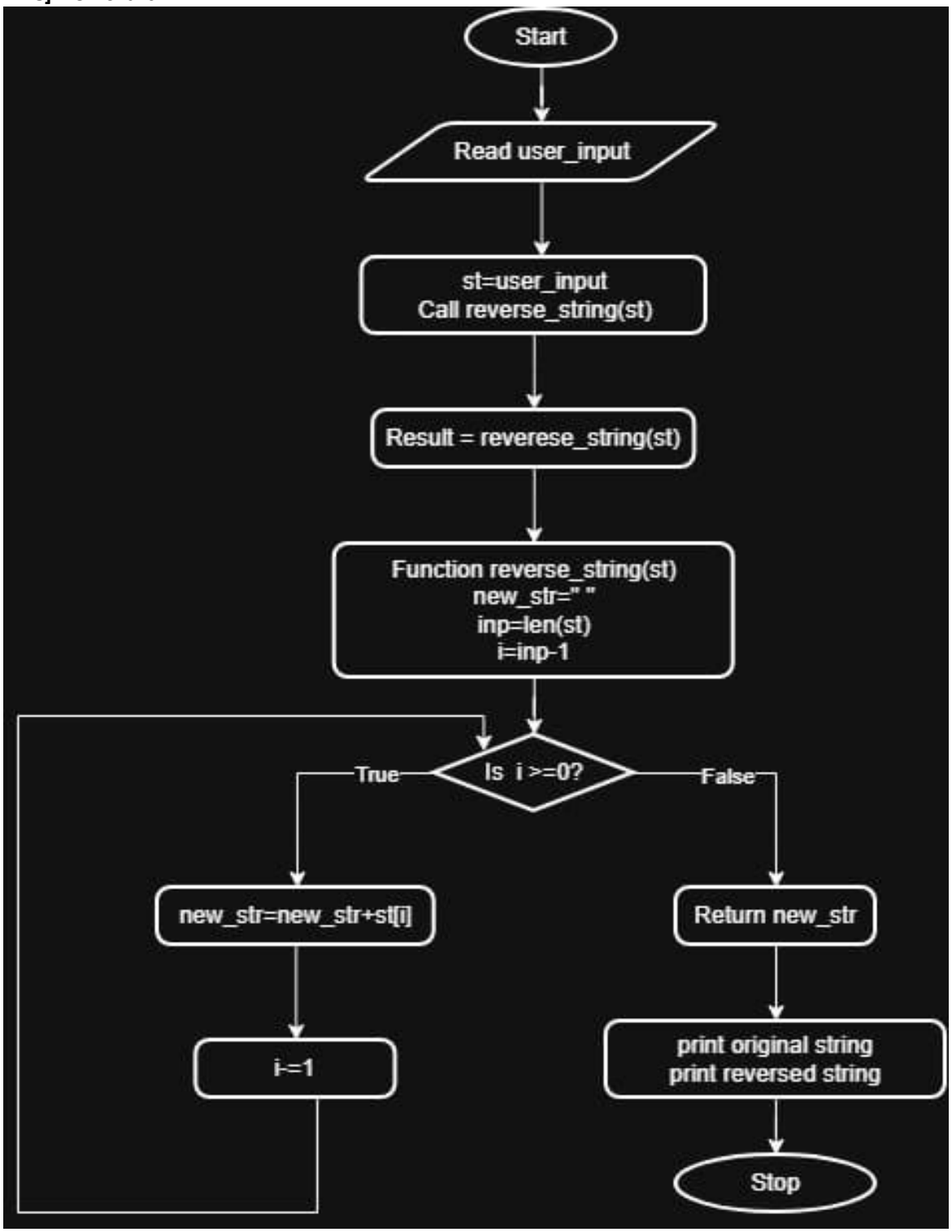
B] code

```
def reverse_string(input_string):
        re=""
        l = len(user_input)
        for i in range(l-1,-1,-1):
                re += user_input[i]
        return re



user_input = input("Enter a string: ")
result = reverse_string(user_input)
print(f"Original String: {user_input}")
print(f"Reversed String: {result}")
```

C] Flowchart


Start
Read user_input
st=user_input
Call reverse_string(st)
Result = reverese_string(st)
Function reverse_string(st)
new_str=" "
inp=len(st)
i=inp-1
Is i >=0?
True
False
new_str=new_str+st[i]
i-=1
Return new_str
print original string
print reversed string
Stop

D] OUTPUT

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10.1.2. Reverse a String

25:48

Write a Python program that prints the reverse of a given input string. The program uses a **reverse_string()** function that takes a string as input and returns its reversed string.

Input Format:

- The program prompts "Enter a string: " and reads a string representing the string to be reversed.

Output Format:

- The first line prints the original string, in the format:

```
Original String: <input_string>
```

- The next line prints the reversed string, in the format:

```
Reversed String: <reversed_string>
```

Sample Test Cases +

Explorer

ReverseS...

Submit

```python
def reverse_string(s):
    return s[::-1]
user_input = input("Enter a string: ")
result = reverse_string(user_input)
print(f"Original String: {user_input}")
print(f"Reversed String: {result}")
```

Debugger

Average time	Maximum time
0.003 s	0.005 s
2.75 ms	5.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 5 ms Debug

Expected output	Actual output
Enter a string: PythonProgrammming	Enter a string: PythonProgrammming
Original String: PythonProgrammming	Original String: PythonProgrammming
Reversed String: gnimmmargorPnohtyP	Reversed String: gnimmmargorPnohtyP

Test case 2 2 ms

Terminal Test cases

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